

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

1		2		3		4		5		6		7		8									
A																							
B																							
C																							
D																							
E																							
F																							
UniSec Medium Voltage Switchgear PT MULTIMAS NABATI ASAHAN MVD-28 20kV Central Office SCHEMATIC DIAGRAM SBC-W OUTGOING H02																							
				Date Prep.		15.11.2018		PROJECT TITLE				ABB		EPMV		COVER SHEET		Scale		NTS		= H02	
				Prepared		YR		MV Switchgear for MNA Serang Plant				PT. ABB Sakti Industri		MVD-28 20kV Central Office		Customer		PT Multimas Nabati Asahan		& 306			
00		Issued for Approval		15.01.19		YR										Checked		AW		Drawing Nbr.		18.042-MNA-306.09	
Ind.		Revision		Date		Name		Approved		AI		Based on: BE318042				SBC-W OUTGOING		Customer Doc. Nbr.				Cont. 17	
1		2		3		4		5		6		7		8									

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

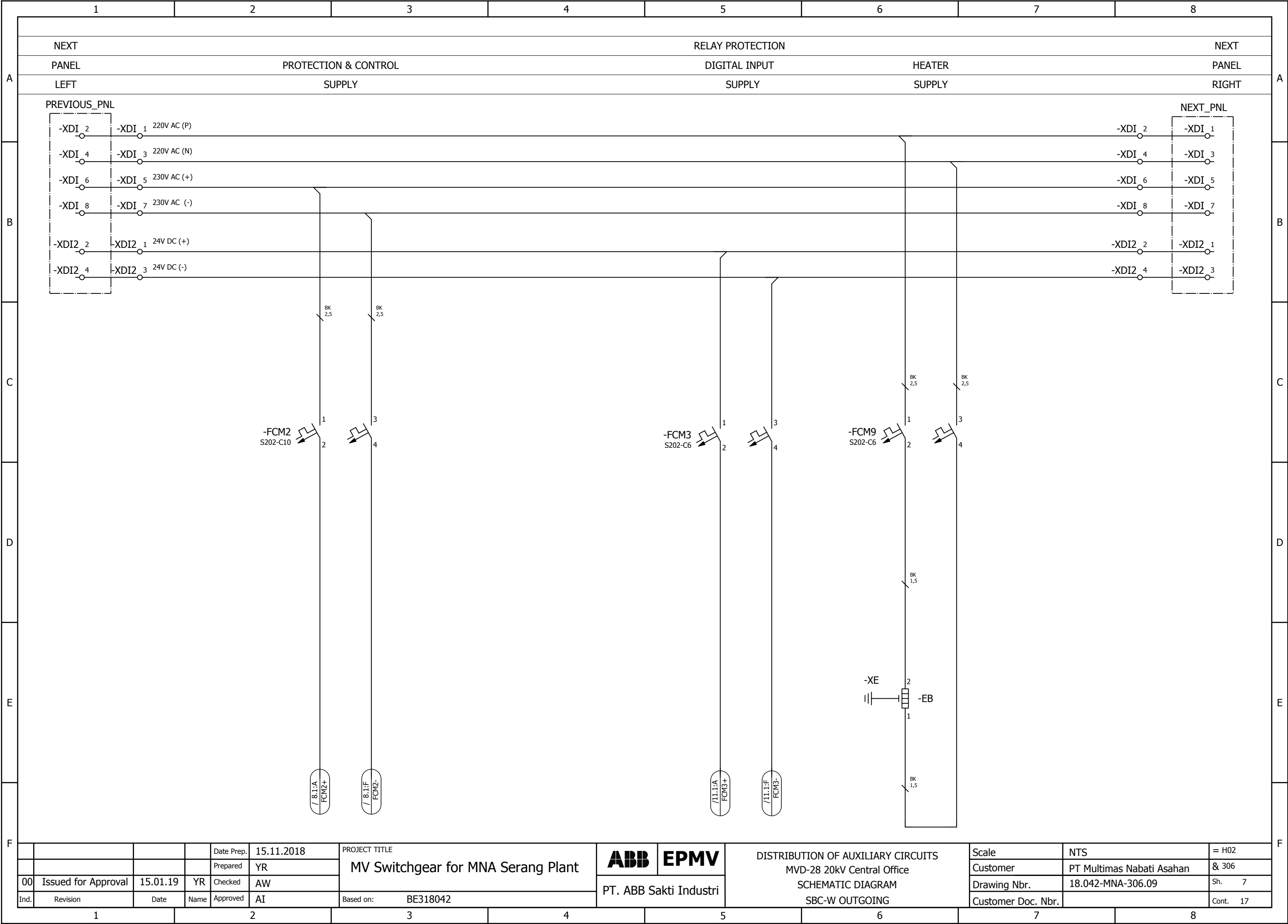
	1	2	3	4	5	6	7	8								
A	SHEET INDEX								A							
	PAGE			PAGE NAME				REVISION								
	==MVD-27=H02&306/1			COVER SHEET				00								
	==MVD-27=H02&306/2			Index of Sheet				00								
	==MVD-27=H02&306/3			REFERENCE DESIGNATIONS				00								
	==MVD-27=H02&306/4			REFERENCE DESIGNATIONS				00								
	==MVD-27=H02&306/5			REFERENCE DESIGNATIONS				00								
	==MVD-27=H02&306/6			REFERENCE DESIGNATIONS				00								
	==MVD-27=H02&306/7			DISTRIBUTION OF AUXILIARY CIRCUITS				00								
	==MVD-27=H02&306/8			AUXILIARY SUPPLY CIRCUIT				00								
		==MVD-27=H02&306/9			CIRCUIT BREAKER CONTROL				00							
==MVD-27=H02&306/10			AUXILIARY MOTOR & RELAY				00									
==MVD-27=H02&306/11			DIGITAL INPUT/OUPUT RELAY				00									
==MVD-27=H02&306/12			SPARE CONTACT				00									
==MVD-27=H02&306/13			CURRENT TRANSFOMER				00									
==MVD-27=H02&306/14			VOLTAGE TRANSFORMER				00									
		==MVD-27=H02&306/15			LV DOOR LAYOUT				00							
	==MVD-27=H02&306/16			LV BOX LAYOUT				00								
	==MVD-27=H02&306/17			PART LIST				00								
	==MVD-27=H02&306/18			Terminal diagram ==MVD-27=H02+-XDA												
	==MVD-27=H02&306/19			Terminal diagram ==MVD-27=H02+-XDB1												
	==MVD-27=H02&306/20			Terminal diagram ==MVD-27=H02+-XDC1												
	==MVD-27=H02&306/21			Terminal diagram ==MVD-27=H02+-XDC2												
	==MVD-27=H02&306/22			Terminal diagram ==MVD-27=H02+-XDI												
	==MVD-27=H02&306/23			Terminal diagram ==MVD-27=H02+-XDI2												
	==MVD-27=H02&306/24			Terminal diagram ==MVD-27=H02+-XDI3												
	==MVD-27=H02&306/25			Terminal diagram ==MVD-27=H02+-XDI6												
	==MVD-27=H02&306/26			Terminal diagram ==MVD-27=H02+-XDV												
	==MVD-27=H02&306/27			Terminal diagram ==MVD-27=H02+-XDX												
										E						
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	Index of Sheet MVD-28 20kV Central Office SCHEMATIC DIAGRAM SBC-W OUTGOING		Scale	NTS	= H02	F
				Prepared	YR	MV Switchgear for MNA Serang Plant		Customer					PT Multimas Nabati Asahan	& 306		
00	Issued for Approval	15.01.19	YR	Checked	AW			PT. ABB Sakti Industri		Drawing Nbr.			18.042-MNA-306.09	Sh. 2		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				Customer Doc. Nbr.		Cont. 17				
	1	2	3	4	5	6	7	8								

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

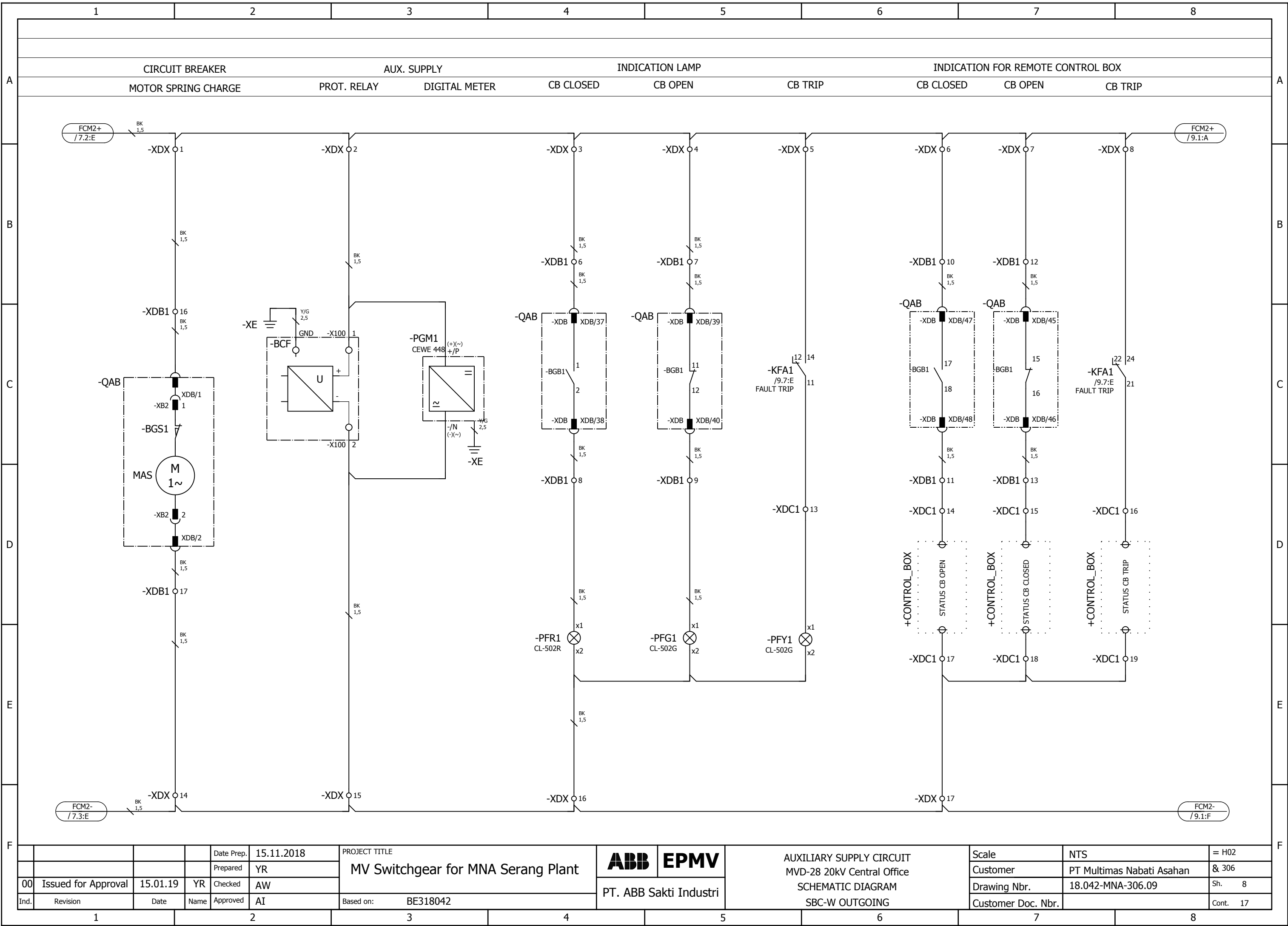
A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS						
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS						
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS						
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS						
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK						
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING						
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH						
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)						
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)						
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES						
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR						
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES						
	-PFB	BLUE SIGNAL LAMPS						-RAI	INDUCTORS						
	-PFF	FLAG RELAYS						-RAR	RESISTORS						
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS						
C	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING						
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT						
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE						
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION						
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD						
	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1						
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2						
	-PFY	YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						
	-PGA	AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS						
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4						
D	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5						
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES						
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS						
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS						
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS						
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS						
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES						
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS						
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS						
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES						
E	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS						
	-PGV	VOLTMETERS						-TB	CONVERTER						
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE						
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE						
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE						
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING						
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT						
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE						
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)													
-QBD4	TWO-WAY DISCONNECTOR														
-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER														
1		2		3		4		5		6		7		8	
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-28 20kV Central Office		Scale	NTS		= H02
				Prepared	YR							Customer	PT Multimas Nabati Asahan		& 306
00	Issued for Approval	15.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM SBC-W OUTGOING		Drawing Nbr.	18.042-MNA-306.09		Sh.
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042					Customer Doc. Nbr.			Cont.

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

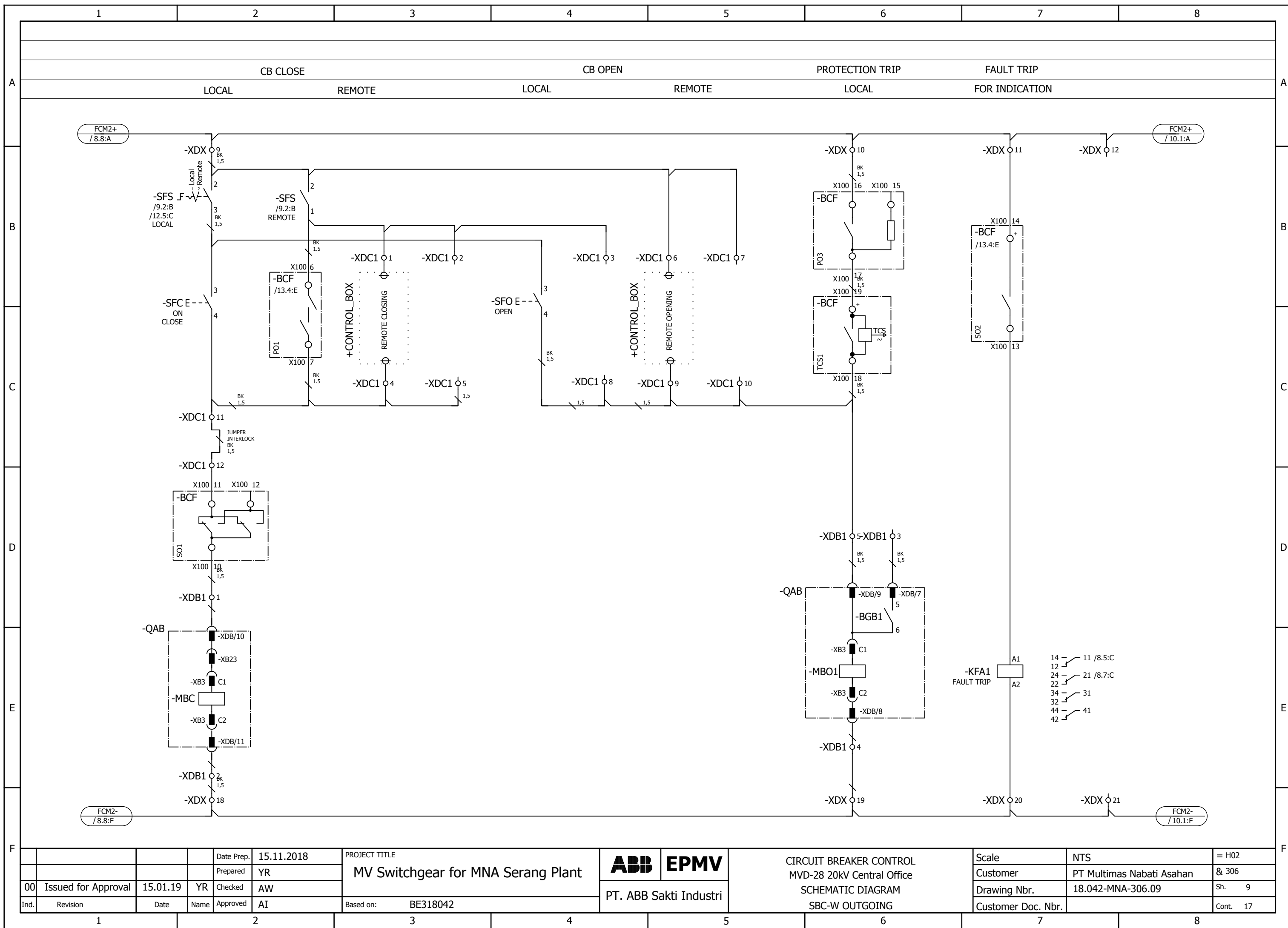
A	-TFA				ACTIVE POWER TRANSDUCERS				-XDI1	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS				A		
	-TFC				CURRENT TRANSDUCERS					-XDI2	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING					
	-TFF				FREQUENCY TRANSDUCERS				-XDI3		TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION					
	-TFJ				ACTIVE ENERGY TRANSDUCERS					-XDI4	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS					
	-TFK				REACTIVE ENERGY TRANSDUCERS				-XDI6		TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS					
	-TFM				MULTIFUNCTION TRANSDUCERS					-XDI7	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS					
	-TFP				POWER-FACTOR TRANSDUCERS				-XDI8		TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS					
	-TFQ				REACTIVE POWER TRANSDUCERS					-XDI9	TERMINAL BLOCK FOR SPECIAL USE					
	-TFS				SIGNAL CONVERTERS				-XDJ		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD					
	-TFV				VOLTAGE TRANSDUCERS					-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS					
B	-WA				BUSBARS				-XDM		SEALABLE TERMINAL BLOCK FOR MEASUREMENT					
	-WBC				POWER CABLES					-XDS	CURRENT SOCKETS					
	-WE				EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.				-XDT		TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB					
	-WGC				CONTROL CABLES					-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK					
	-WGS				COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)				-XDU1		TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK					
	-WH				OPTICAL FIBRES					-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS					
	-XDA				TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS				-XDV1		CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1					
	-XDA1				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1					-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2					
	-XDA2				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2				-XDV3		CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3					
	-XDA3				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3					-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR					
	-XDA5				CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER				-XDX		SUPPORT TERMINAL BLOCK					
	-XDB				CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS					-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE					
	-XDB1				CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)				-XE2		EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE					
	-XDB2				CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER					-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE					
	-XDB10...-XDB89				CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS				-XE6		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE					
	-XDB7				CONNECTOR FOR TRUCK POSITION CONTACTS					-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE					
	-XDB91				CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR				-XE8		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE					
	-XDB92				CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR					-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT					
	-XDB93				CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR											
	-XDB95				CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET											
D	-XDC				CUSTOMER TERMINAL BLOCK											
	-XDC1				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER											
	-XDC2				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR											
	-XDC3				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT											
	-XDC4				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS											
	-XDC5				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER											
	-XDE				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE											
	-XDE1				TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE											
	-XDE2				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2											
	-XDE3				TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2											
	-XDF				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS											
	-XDF1				CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR											
	-XDF2				CONNECTOR FOR REAR VENTILATOR CIRCUITS											
	-XDF3				CONNECTOR FOR FRONT VENTILATOR CIRCUITS											
	-XDH				TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)											
	-XDI				TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)											
F					Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB		EPMV	REFERENCE DESIGNATIONS MVD-28 20kV Central Office SCHEMATIC DIAGRAM SBC-W OUTGOING		Scale	NTS	
					Prepared	YR				Customer				PT Multimas Nabati Asahan		& 306
	00	Issued for Approval	15.01.19	YR	Checked	AW				Drawing Nbr.				18.042-MNA-306.09		Sh. 6
	Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042		PT. ABB Sakti Industri		Customer Doc. Nbr.			Cont. 17	
1		2		3		4		5		6		7		8		



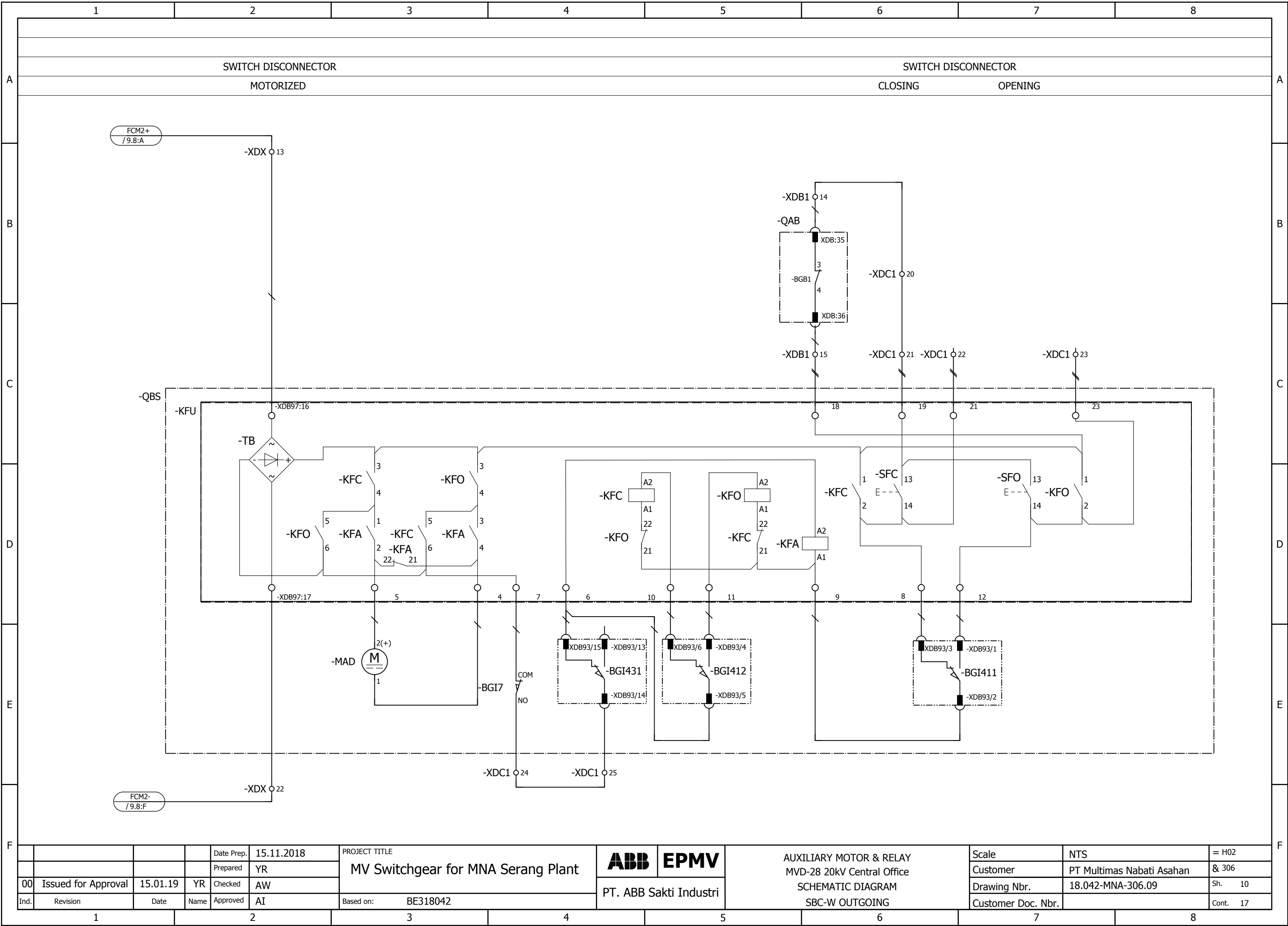
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



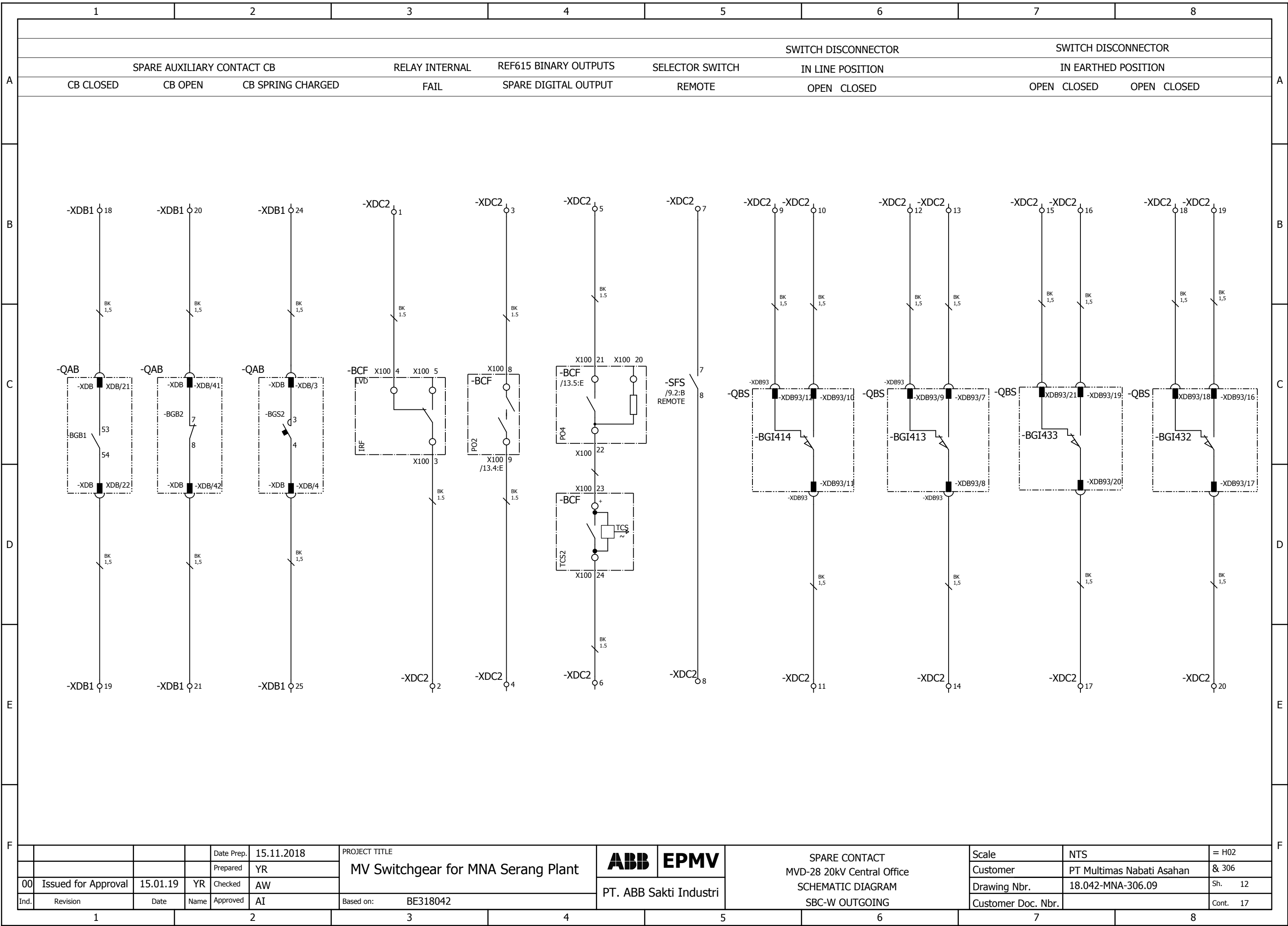
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



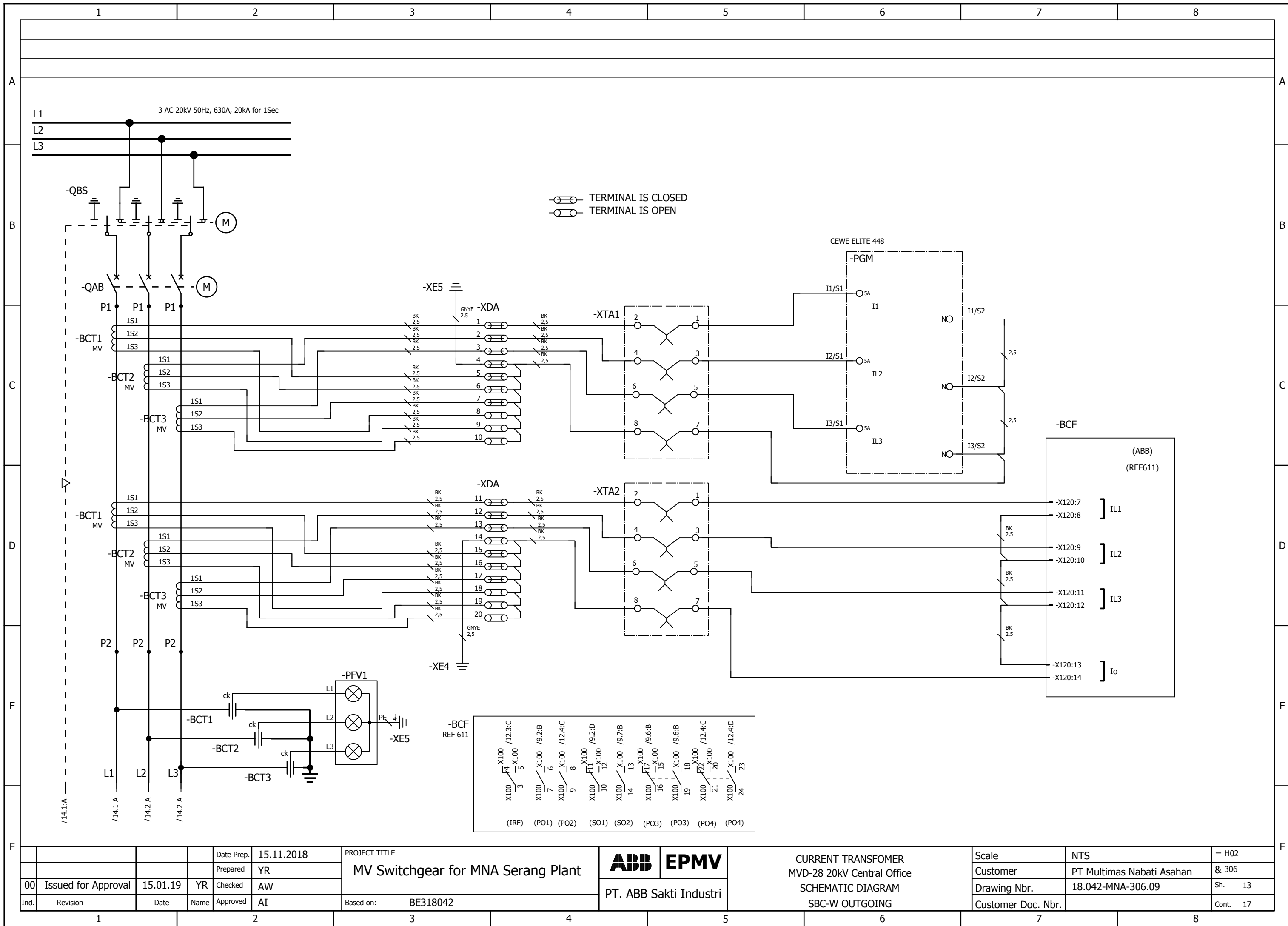
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

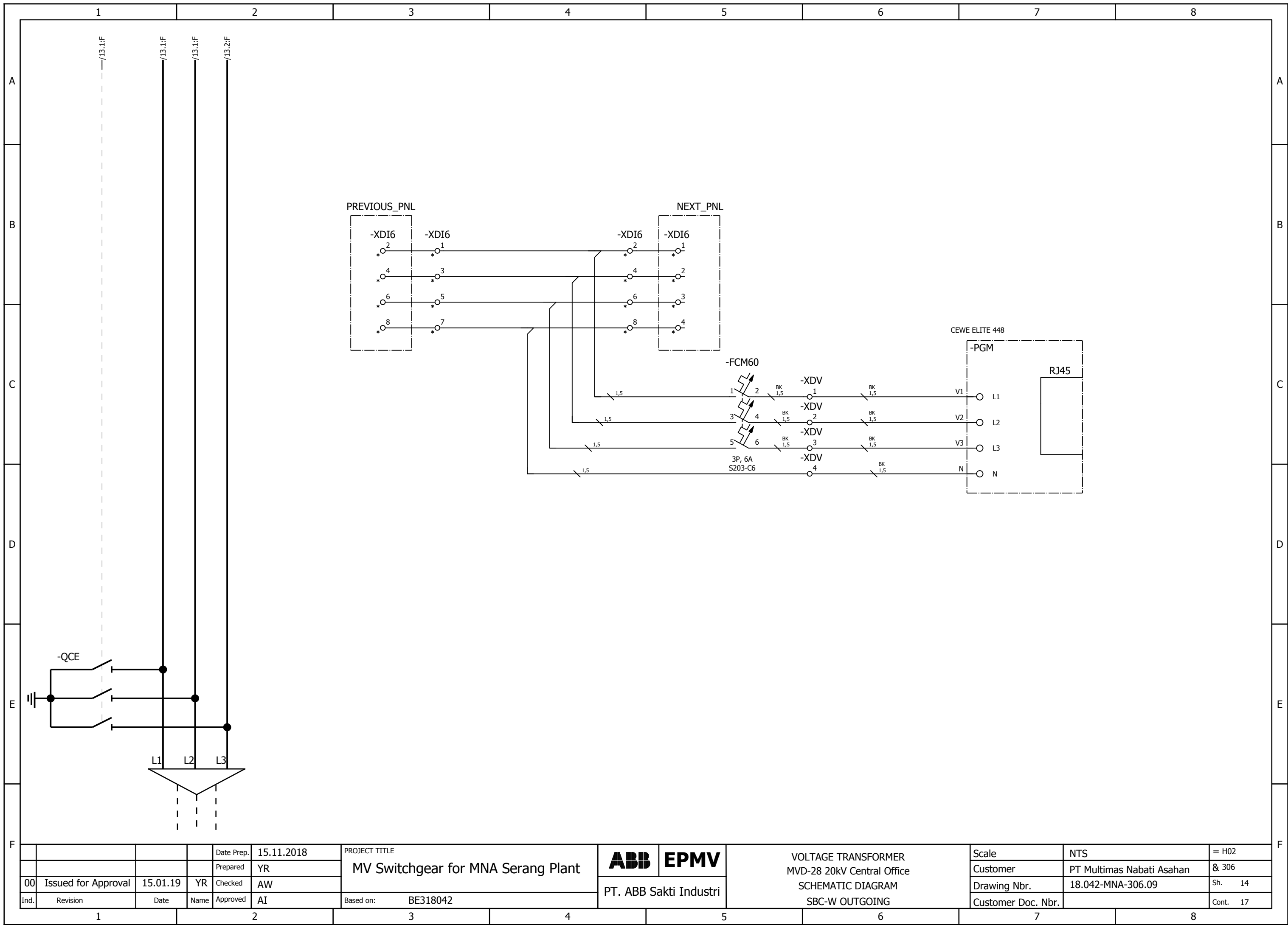


WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.





No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK

[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-28 20kV Central Office SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	15.01.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		18.042-MNA-306.09	Sh.	16
Ind.	Revision	Date	Name	Approved	AI	Based on:		BE318042		Customer Doc. Nbr.		Cont.

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

	1	2	3	4	5	6	7	8
A	Summarized parts list							
	IDABB_F02_005-IDABB							
	Carry over: 0.00							
	TAG ID	DESCRIPTION	MANUFACTUR/BRAND	PART NUMBER	QUANTITY	REMARKS		
	-BCF	Feeder Management Relay REF611HBAABC1NN11G	ABB	REF611HBAABC1NN11G	1	/8.2:C		
	-EB	HEATER 230VAC, 50W	SANYANG	CDXN010033P0009	1	/7.6:E		
	-FCM2	MCB S202-C10 MCB. 2P, 10A	ABB	2CDS252001R0104	1	/7.2:C		
	-FCM3;-FCM9	MCB S202-C6 MCB. 2P, 6A	ABB	2CDS252001R0064	2	/7.5:C;/7.6:C		
	-FCM60	MCB S203-C6 MCB. 3P, 6A	ABB	2CDS253001R0064	1	/14.5:C		
	-KFA1	Auxiliary Relay 230AC 4COS	PHOENIX	1YRA3DL266230001	1	/9.7:E		
B	-PFG1	PILOT LIGHT GREEN #CL-523G#230VAC CL-523G	ABB	1SFA619402R5232	1	/8.5:E		
	-PFR1	PILOT LIGHT RED #CL-523R#230VAC CL-523R	ABB	1SFA619402R5231	1	/8.4:E		
	-PFY1	PILOT LIGHT YELLOW #CL-523Y#230VAC CL-523Y	ABB	1SFA619402R5233	1	/8.6:E		
	-PGM1	POWER METER CEWE 448 Class : 0.5s; CT Input : 1A; VT Input : 57.7-240V (L-N); Aux. Voltage: 80-300VAC/DC	CEWE	1YRA4EA0350001	1	/8.3:C		
	-SFC	PUSH BUTTON, YW1B-M1E10-G GREEN COLOUR	IDEC	1YRA3DA14700001	1	/9.2:B		
C	-SFO	PUSH BUTTON, YW1B-M1E10-R RED COLOUR	IDEC	1YRA3DA14700002	1	/9.4:B		
	-SFS	LOCAL/REMOTE Selector Switch Type : S0-CA10-A723-600-E; 4NC (for LOCAL Position) & 4NO (for REMOTE Position)	K&N	1YRA3DA17940011	1	/9.2:B		
	-XTA1;-XTA2	Current Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting	YONGSUNG	YSCTT-04C	2	/13.4:C;/13.4:D		
D								
E								
F								

				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	PART LIST		Scale	NTS	= H02	
				Prepared	YR	MV Switchgear for MNA Serang Plant				MVD-28 20kV Central Office		Customer	PT Multimas Nabati Asahan	& 306	
00	Issued for Approval	15.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-306.09	Sh. 17	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				SBC-W OUTGOING		Customer Doc. Nbr.		Cont. 17	
1		2		3		4		5		6		7		8	